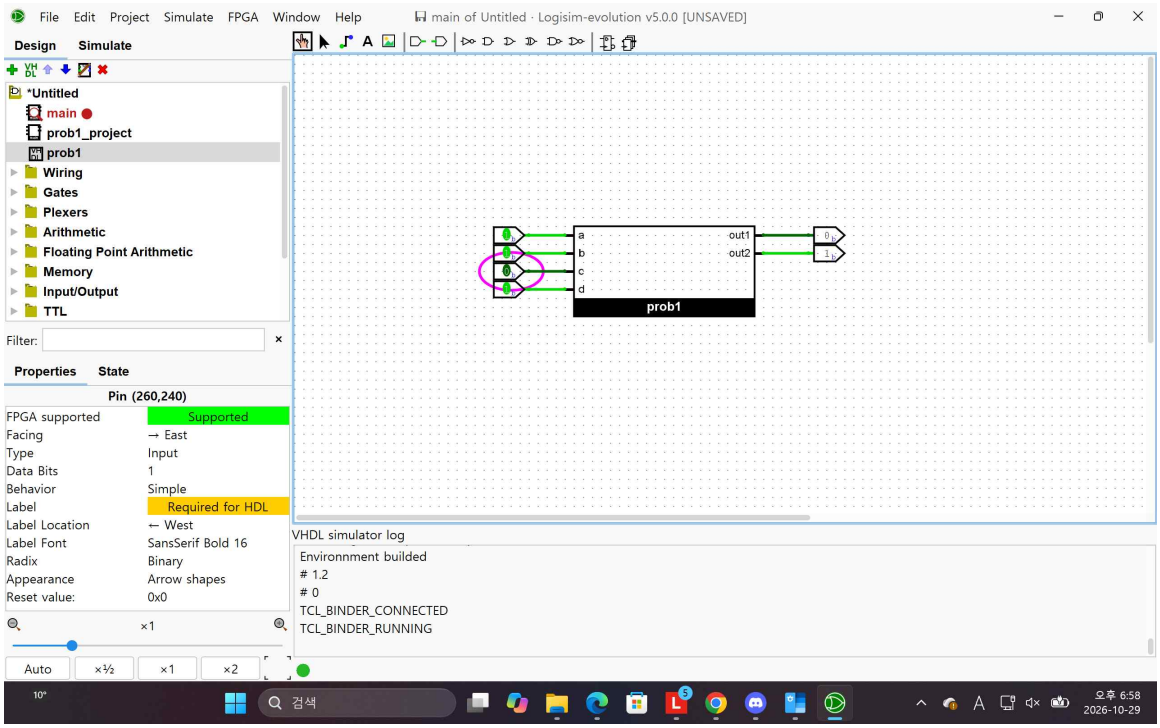
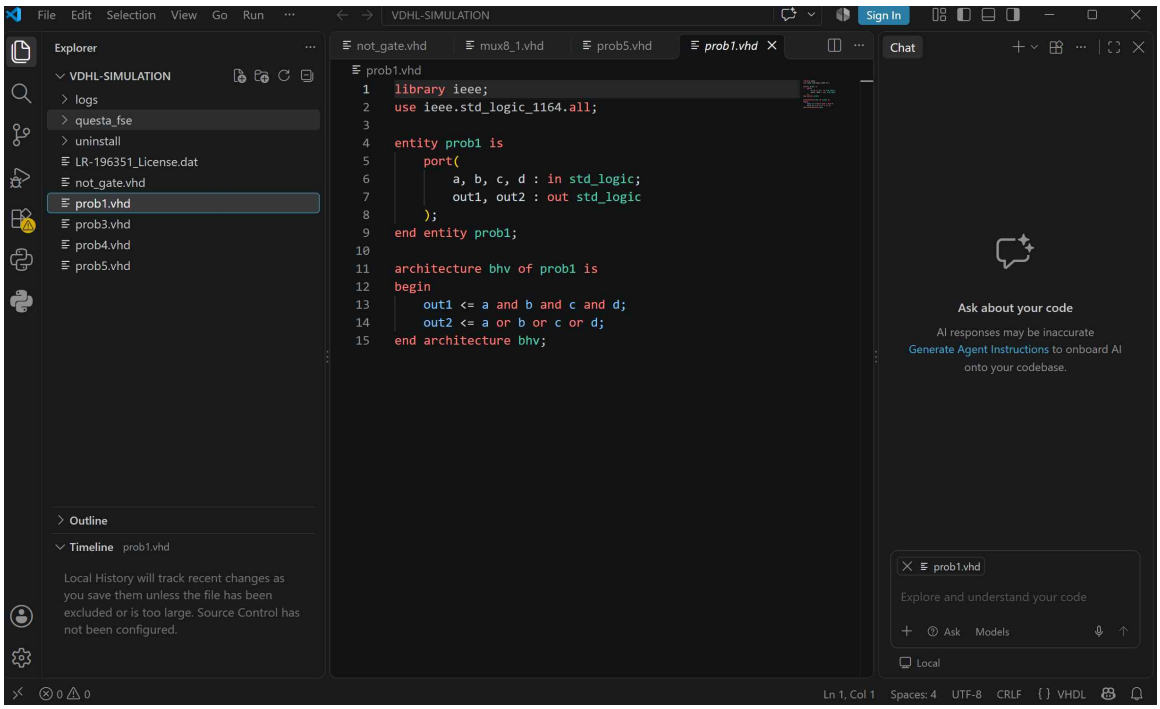
1. Write VHDL code.

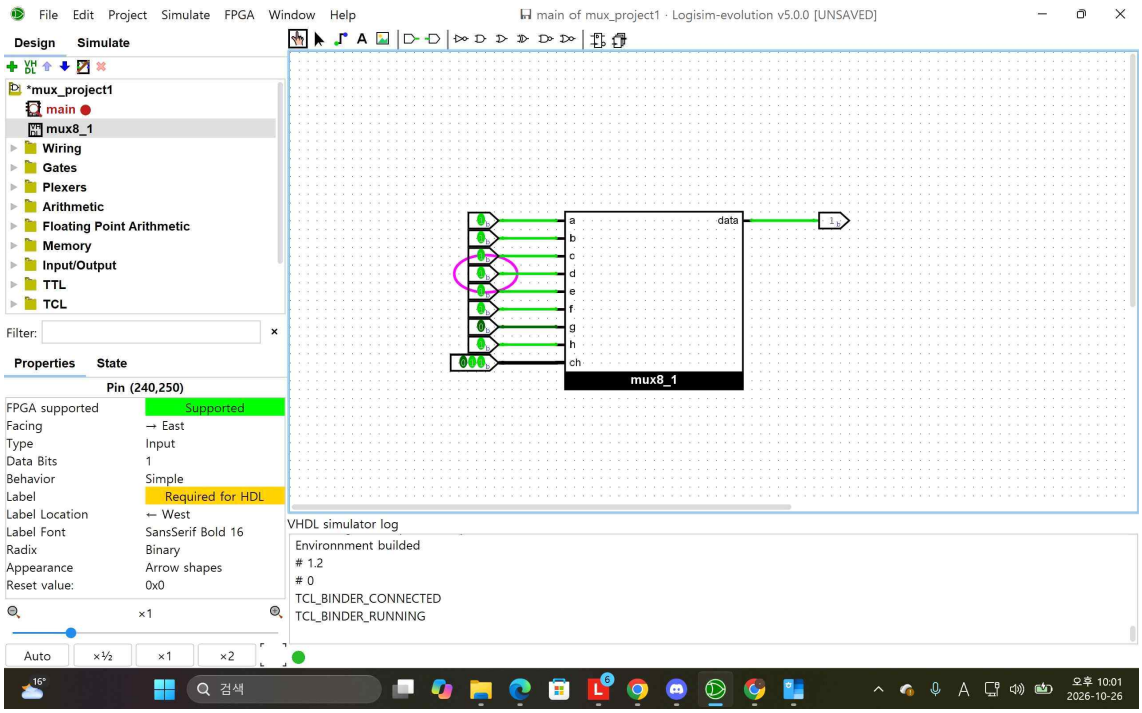


VDHL

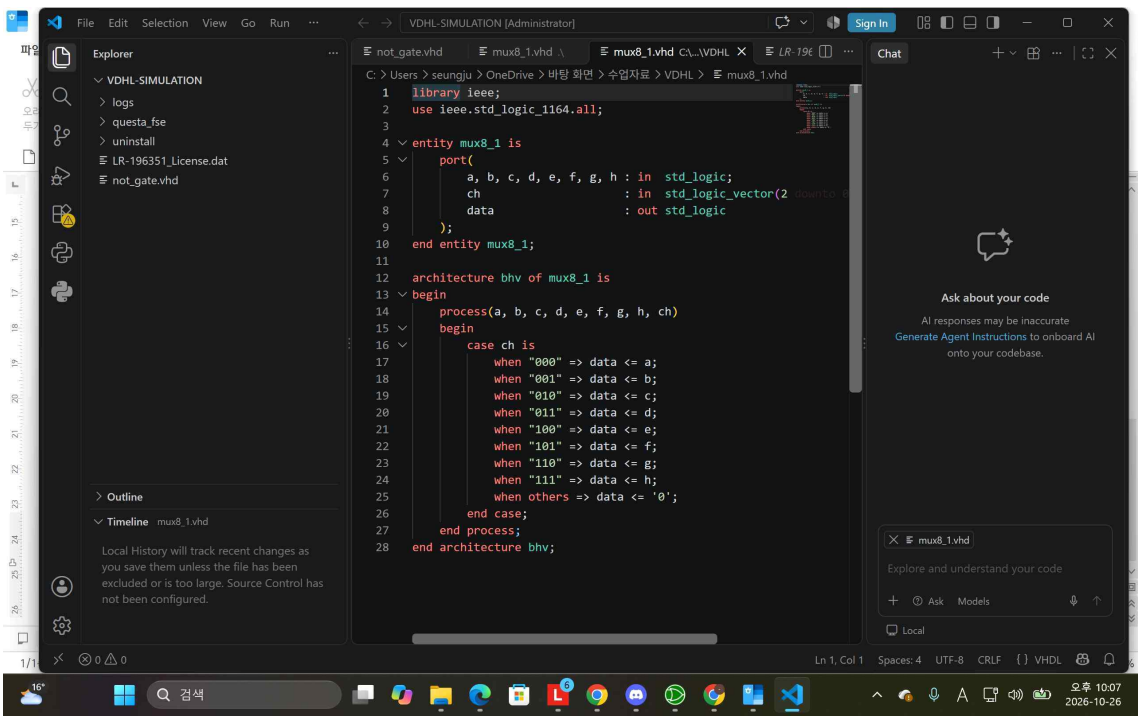


VS CODE

2. 8-to-1 Multiplexer

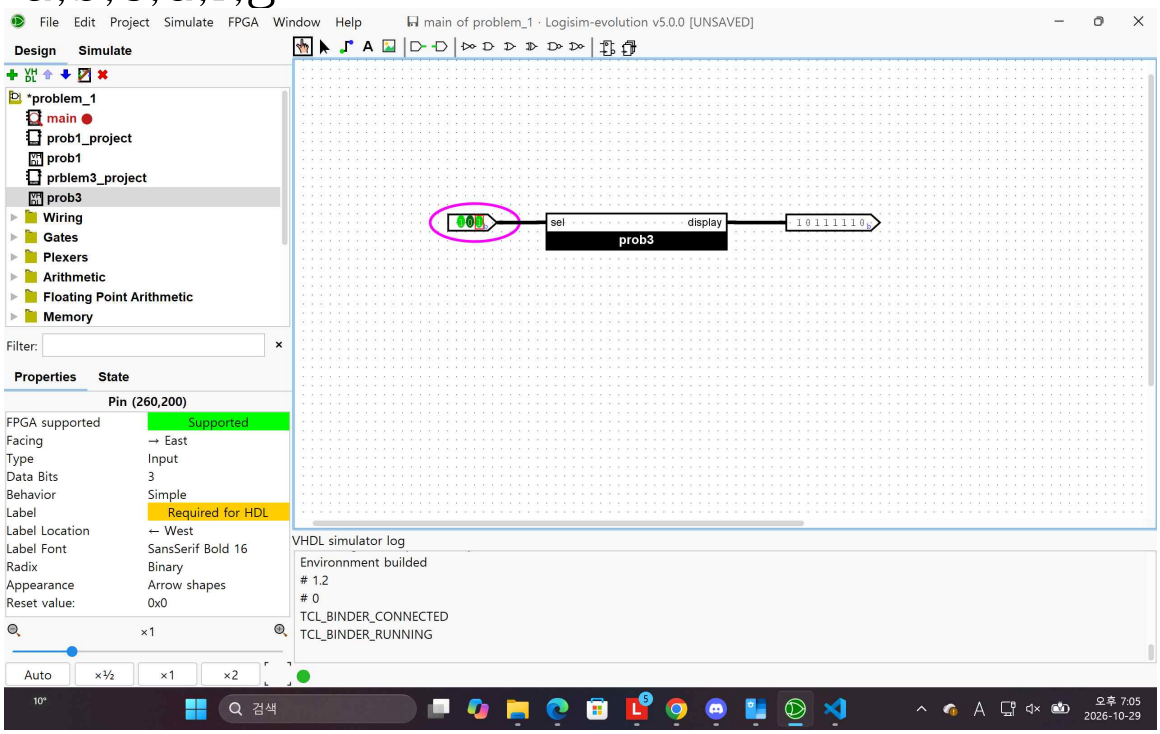


VHDL

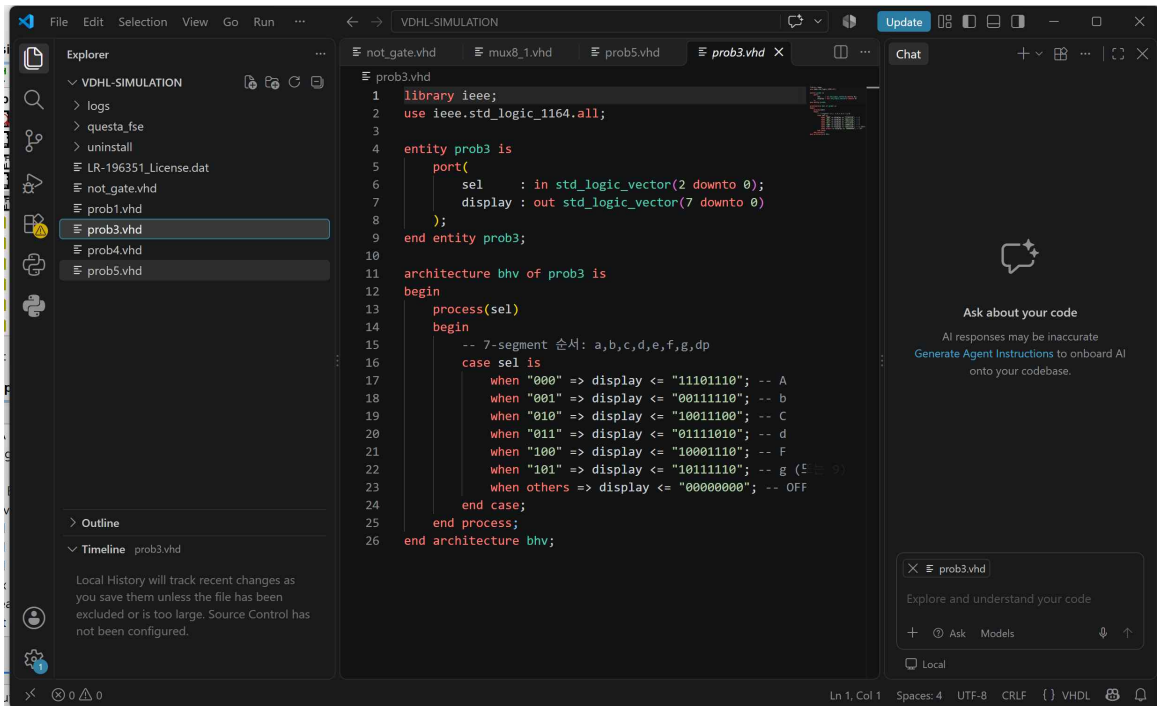


VS CODE

3.Use digital tubes to display letters: a,b,c,d,f,g



VDHL



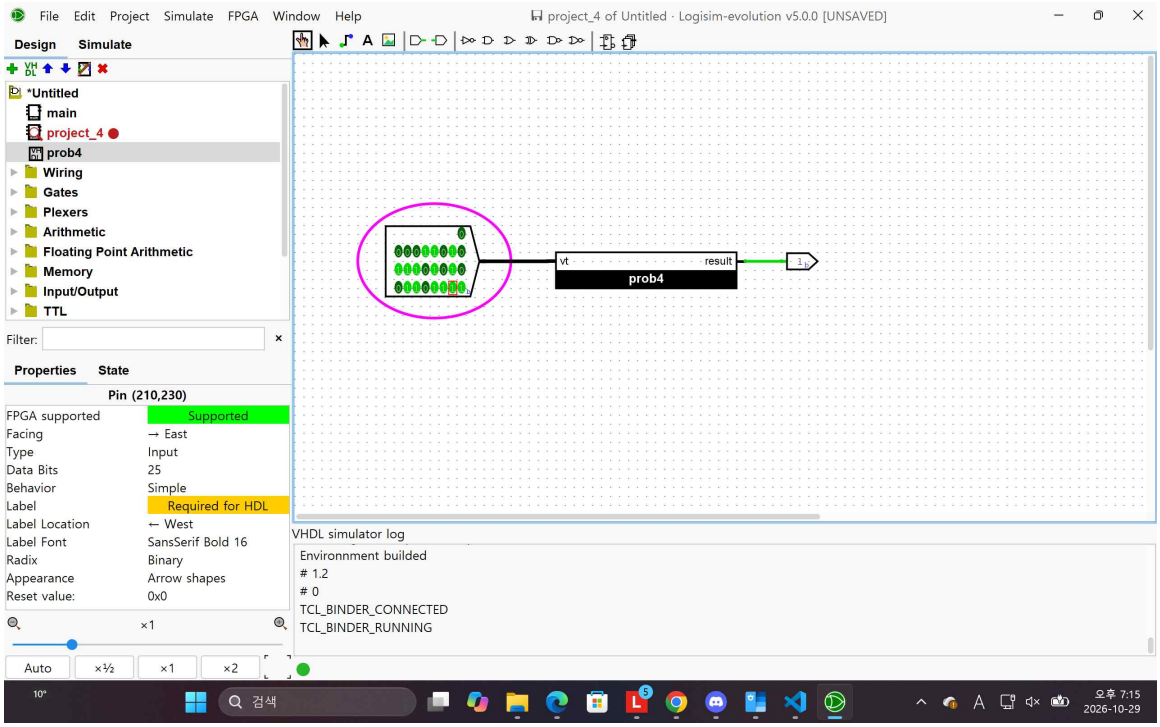

```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity prob3 is
    port(
        sel     : in std_logic_vector(2 downto 0);
        display : out std_logic_vector(7 downto 0)
    );
end entity prob3;

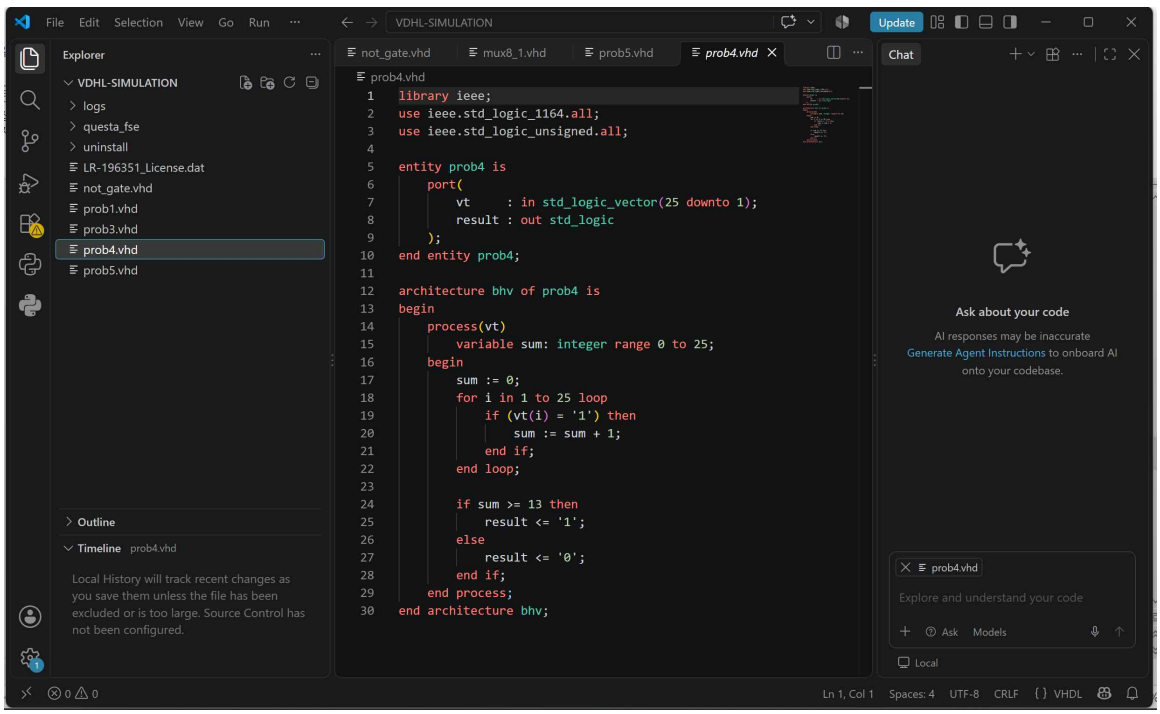
architecture bhv of prob3 is
begin
    process(sel)
    begin
        -- 7-segment 순서: a,b,c,d,e,f,g,dp
        case sel is
            when "000" => display <= "11101110"; -- A
            when "001" => display <= "00111110"; -- b
            when "010" => display <= "10011100"; -- C
            when "011" => display <= "01111010"; -- d
            when "100" => display <= "10001110"; -- F
            when "101" => display <= "10111110"; -- g (5
            when others => display <= "00000000"; -- OFF
        end case;
    end process;
end architecture bhv;
```

VS CODE

4. 25-persion voting machine



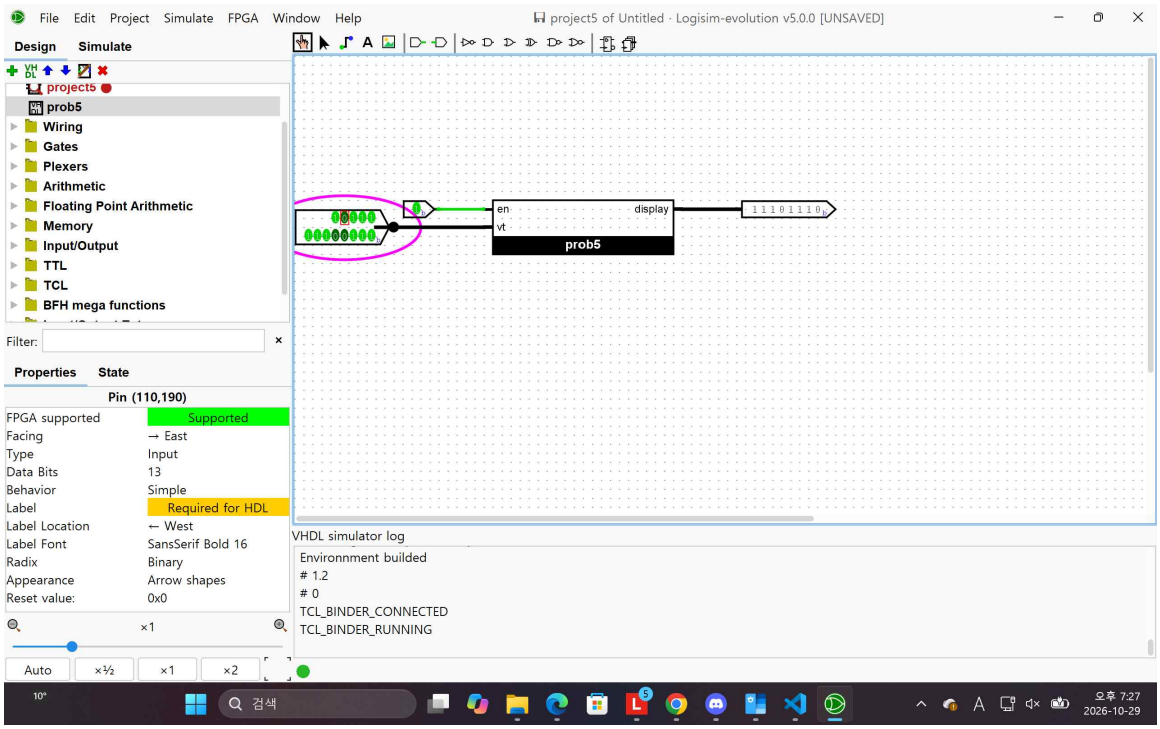
VDHL



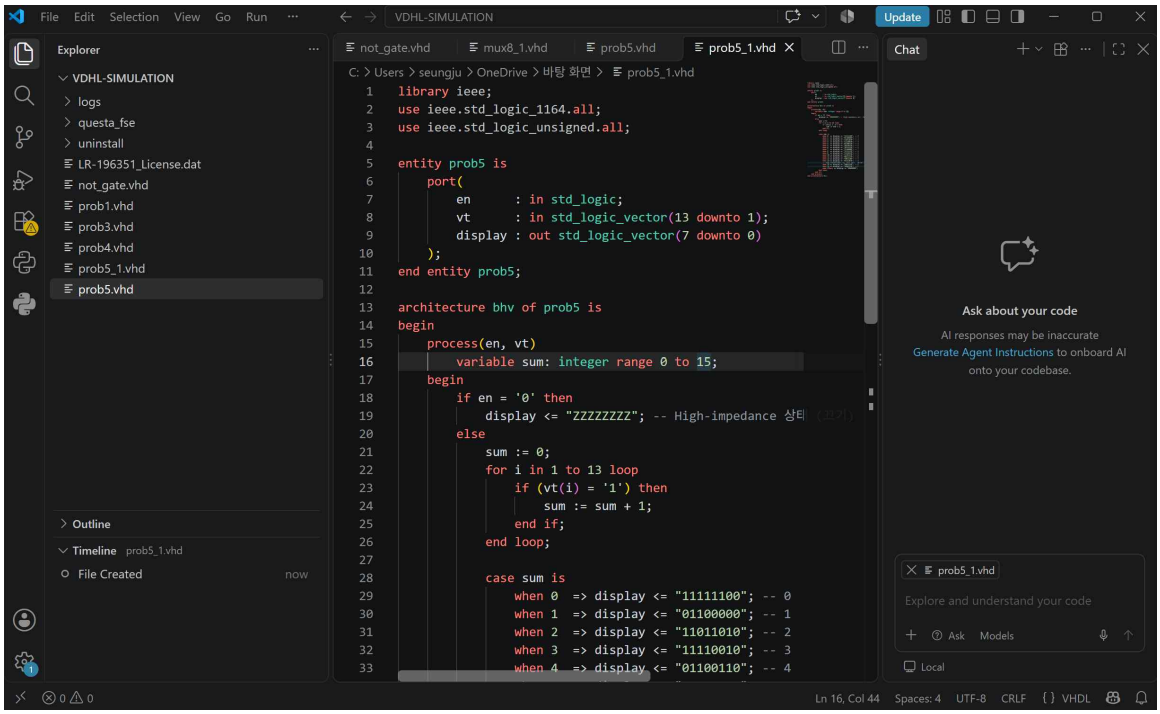

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3  use ieee.std_logic_unsigned.all;
4
5  entity prob4 is
6      port(
7          vt     : in std_logic_vector(25 downto 1);
8          result : out std_logic
9      );
10 end entity prob4;
11
12 architecture bhv of prob4 is
13 begin
14     process(vt)
15         variable sum: integer range 0 to 25;
16     begin
17         sum := 0;
18         for i in 1 to 25 loop
19             if (vt(i) = '1') then
20                 sum := sum + 1;
21             end if;
22         end loop;
23
24         if sum >= 13 then
25             result <= '1';
26         else
27             result <= '0';
28         end if;
29     end process;
30 end architecture bhv;
```

VS CODE

5.



VDHL



VS CODE